

3GEM POOLING: DIRECTION-AWARE AND COMPACT GLOBAL DESCRIPTORS FOR VISUAL PLACE RECOGNITION

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ABSTRACT

Visual Place Recognition (VPR) is crucial for applications such as augmented reality (AR) and virtual reality (VR), where lightweight architectures and compact global descriptors enable practical deployment. Generalized Mean (GeM) pooling is widely used for global aggregation due to its simplicity and effectiveness, but its uniform spatial averaging treats all local features equally, often incorporating non-discriminative regions like pedestrians or sky. To overcome this, we propose 3GeM Pooling, which aggregates along both spatial axes and the channel dimension, capturing richer directional cues. We further introduce a Spatially Guided Channel Enhancement (SGCE) module that leverages spatially pooled information to reinforce channel-wise aggregation, producing more compact and discriminative descriptors. This design enables finer modeling of spatial structures while maintaining GeM's lightweight nature. With equal dimensionality, 3GeM improves Recall@1 by 7.8% over GeM on MSLS-val and achieves state-of-the-art results on multiple challenging VPR benchmarks.

Index Terms— deep learning, visual place recognition, pooling method

1. INTRODUCTION

Visual Place Recognition (VPR) estimates the approximate location of a query image by retrieving visually similar references from a georeferenced database [1, 2, 3]. It plays a crucial role in augmented reality (AR) [4], autonomous navigation, and SLAM-based robot localization [5]. Despite substantial progress, VPR remains challenging because the same place may vary widely with illumination, weather, or viewpoint, while distinct places may look deceptively similar (perceptual aliasing) [6]. These difficulties are critical when methods rely on compact global descriptors that must be both highly discriminative and efficient.

A common approach views VPR as image retrieval: encode query and database images into compact global descriptors and compare them for nearest-neighbor search [2, 7, 8]. Retrieval-based methods are either one-stage, which match using global descriptors, or two-stage, which shortlist candidates via global descriptors and then refine them with local matching or geometric verification [9, 10]. In both cases, performance relies on descriptors that are robust to viewpoint and appearance changes yet small enough for fast retrieval and storage [8]. Among global aggregation methods, NetVLAD [2] selectively aggregates local features via soft assignment to learned clusters, but often yields high-dimensional descrip-

tors. GeM pooling [7] is simpler and compact, averaging spatial responses with a generalized mean. However, GeM collapses spatial variation into a uniform average, overlooking directional structures (e.g., vertical facades vs. horizontal roads) that are critical in VPR. This motivates us to design a pooling strategy that explicitly encodes directional cues.

We address this with Direction-aware Generalized Mean pooling (3GeM), which performs GeM along three axes—channel, horizontal (width), and vertical (height)—to capture complementary directional structure that standard spatial pooling misses. We further introduce a Spatially Guided Channel Enhancement (SGCE) module that leverages spatially pooled signals to reweight and refine channel-wise aggregation, producing compact yet more discriminative descriptors while preserving GeM's lightweight nature. Unlike cluster-based methods such as SuperVLAD, which trade efficiency for discriminative power, our approach retains the compactness of GeM while introducing directional aggregation, achieving state-of-the-art results with only 1024-D descriptors.

Contributions. (i) We propose 3GeM, a direction-aware pooling strategy along channel/height/width that enriches global descriptors without codebook assignment. (ii) We design SGCE, a spatially guided channel enhancer that integrates directional cues before channel fusion. (iii) Our method retains GeM's interface, enabling easy integration into existing pipelines without backbone changes. Despite its plug-and-play design and compact output, it surpasses prior SOTA on MSLS-val, MSLS-challenge, and Nordland, showing strong generalization across diverse conditions.

2. RELATED WORK

From handcrafted features to deep descriptors. Early VPR relied on local features such as SIFT [11], SURF [12], and ORB [13], aggregated with BoW [14] or VLAD [15]. While effective in controlled settings, these pipelines degrade under large viewpoint or appearance changes.

Global aggregation: NetVLAD vs. GeM. NetVLAD [2] learns soft assignment to cluster centers, enabling selective aggregation but typically producing high-dimensional descriptors. GeM pooling [7] aggregates spatial responses with a generalized mean, yielding compact descriptors popular for real-time deployment, yet its uniform spatial averaging can admit non-discriminative regions. Our work builds on this idea and improves the aggregation method of GeM. We keep the descriptor compact while making it more discriminative.

Recent advances and foundation backbones. Large-scale place-centric datasets (e.g., GSV-Cities and SF-XL) and stronger backbones have driven progress in both one- and two-stage systems. CNN-based CosPlace [16] and EigenPlaces [17] reformulate training as place classification, while MixVPR [18] mixes CNN features with MLPs and metric learning to improve discrimination. Two-

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stage methods (e.g., TransVPR, SelaVPR) refine global candidates via local/geometric verification [9, 1]. With ViT backbones, self-supervised DINOv2 [19] has been widely adapted to VPR; SALAD and FoL-global incorporate DINOv2 features for robust urban retrieval [20, 10], and CricaVPR models cross-image correlations on DINOv2 features [21].

Lightweight, deployment-friendly descriptors. There is a growing trend toward compact descriptors that retain robustness while enabling fast retrieval at scale. For example, Lu et al. propose implicit aggregation (ImAge), which adds learnable aggregation tokens to a Transformer backbone to form the global descriptor via self-attention, avoiding an explicit aggregator [22]. SuperVLAD [8] reduces the number of cluster centers used in NetVLAD while still maintaining competitive performance. Our approach follows this trend by preserving GeM’s compactness yet enhancing discrimination via direction-aware pooling (3GeM) and spatially guided channel fusion (SGCE), which together improve robustness without increasing descriptor size.

3. METHOD

Our method uses a ViT backbone (DINOv2 [19]) with GeM pooling [7] as baseline. We extend it to 3GeM pooling, which aggregates features along three axes, and introduce SGCE to refine channel descriptors with spatial cues. (see Fig. 1)

3.1. 3GeM Pooling Layer

Given an input image, a ViT backbone produces a feature map $X \in \mathbb{R}^{H \times W \times D}$ in addition to a class token. While the class token can serve as a global descriptor, we focus on pooling the feature map. Standard GeM pooling [7] computes each channel descriptor as

$$G_d[k] = \left(\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W x_{i,j,k}^{p_d} \right)^{1/p_d}, \quad k = 1, \dots, D, \quad (1)$$

where p_d is a learnable pooling parameter that interpolates between average ($p_d = 1$) and max pooling ($p_d \rightarrow \infty$). Although effective, GeM pooling pools uniformly over $H \times W$, yielding descriptors that may miss directional or localized cues important for VPR.

To address the limitation of GeM pooling which aggregates only over $H \times W$ and thus may overlook directional cues, we propose **3GeM pooling**. Given a feature map $X \in \mathbb{R}^{H \times W \times D}$, 3GeM applies GeM along three orthogonal dimensions: $H \times W$, $H \times D$, and $W \times D$. In addition to the channel-wise descriptor G_d defined in (1), we compute

$$G_w[j] = \left(\frac{1}{HD} \sum_{i=1}^H \sum_{k=1}^D x_{i,j,k}^{p_w} \right)^{1/p_w}, \quad j = 1, \dots, W, \quad (2)$$

$$G_h[i] = \left(\frac{1}{WD} \sum_{j=1}^W \sum_{k=1}^D x_{i,j,k}^{p_h} \right)^{1/p_h}, \quad i = 1, \dots, H. \quad (3)$$

The three descriptors $G_d \in \mathbb{R}^D$, $G_w \in \mathbb{R}^W$, and $G_h \in \mathbb{R}^H$ capture complementary channel-, width-, and height-wise information. They are concatenated and forwarded to the SGCE module for further fusion.

3.2. Spatially Guided Channel Enhancement module

The Spatially Guided Channel Enhancement (SGCE) module is designed to leverage spatial information from height (H) and width (W) dimensions to enhance channel-wise (D) feature discriminability in VPR (see Fig. 2). Traditional feature fusion methods often treat spatial and channel features as separate entities, leading to suboptimal integration of structural cues. SGCE addresses this limitation through a three-stage process:

Spatial Attention Generation. The module first concatenates height-wise features G_h and width-wise features G_w into a unified spatial representation $G_s = [G_h || G_w] \in \mathbb{R}^{H+W}$. This combined feature passes through a compact attention generator comprising two linear layers with ReLU and Sigmoid activations. Crucially, this transforms low-dimensional spatial information into a channel-aligned attention map $\text{Attention}_s \in \mathbb{R}^D$,

$$\text{Attention}_s = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot G_s)), \quad (4)$$

where $W_1 \in \mathbb{R}^{\frac{D}{4} \times (H+W)}$, $W_2 \in \mathbb{R}^{D \times \frac{D}{4}}$, and each element indicates the spatial importance of a specific channel.

Channel Feature Enhancement. The generated spatial attention weights are applied to the original channel features via element-wise multiplication:

$$\text{Attended}_c = G_d \odot \text{Attention}_s, \quad (5)$$

where $\odot$ denotes element-wise multiplication. This operation selectively amplifies channel components aligned with salient spatial regions. The attended features Attended_c then undergo nonlinear enhancement through a bottleneck MLP with GELU activation to obtain the enhanced features:

$$\text{Enhanced}_c = W_4 \cdot \text{GELU}(W_3 \cdot \text{Attended}_c). \quad (6)$$

where $W_3, W_4 \in \mathbb{R}^{D \times D}$. This expansion-reduction structure introduces feature abstraction while preserving dimensionality, effectively encoding spatial-contextual relationships into channel features.

Adaptive Feature Fusion. To maintain original feature integrity, SGCE employs a learnable gating mechanism:

$$G = (1 - \gamma) \cdot G_d + \gamma \cdot \text{Enhanced}_c, \quad (7)$$

where γ is a trainable parameter that adaptively balances raw channel features and spatially enhanced features, and the global descriptor $G \in \mathbb{R}^D$ denotes the final D-dimensional channel-wise summary. This residual formulation improves robustness to noisy spatial inputs while preserving gradient flow through both branches.

4. EXPERIMENTS

4.1. Datasets and Performance Evaluation

We evaluate our method on standard VPR benchmarks covering diverse challenges such as viewpoint, illumination, weather, and seasonal changes. The evaluation includes Pitts30k, MSLS (validation and challenge sets), and Nordland. We report recognition performance using Recall@N (R@N), the fraction of queries whose top- N retrievals contain at least one correct match [23, 24, 25]. A match is defined within 25 m for Pitts30k, within 25 m and 40° orientation for MSLS, and within ± 10 frames for Nordland.

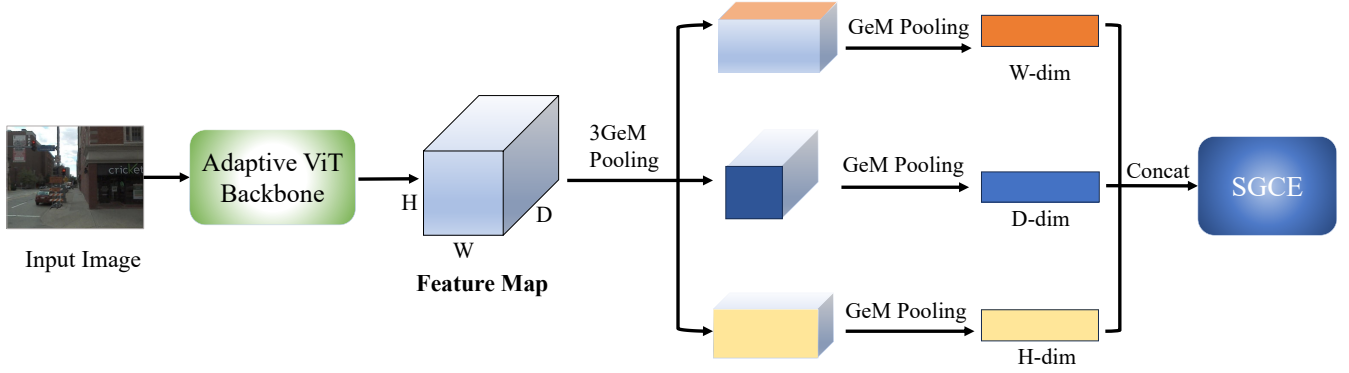


Fig. 1. Illustration of the proposed 3GeM Pooling layer. It performs GeM Pooling along three directions of the feature map generated by the backbone, and then integrates spatial features into the channel-wise representation through the SGCE module.

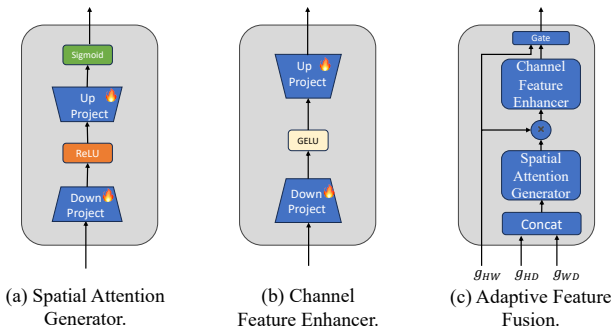


Fig. 2. Illustration of the proposed SGCE module. It includes a spatial attention generator and a Channel Feature Enhancer, which enhances channel features using spatial information.

4.2. Implementation Details

All experiments are implemented in PyTorch with DINOv2 (base/large) backbones and trained on RTX 3090 GPUs. For SOTA comparisons we use 322×322 images, while ablations are trained at 224×224 and inferred at 322×322 . We freeze most of the backbone and fine-tune the last 5 transformer blocks together with our modules. Training uses AdamW (lr=1e-5, weight decay=1e-2, cosine schedule), batch size 64, for 10 epochs. Multi-Similarity loss [26] is adopted to supervise descriptor learning.

4.3. Comparison with State-of-the-Art Methods

We compare the proposed 3GeM Pooling with several state-of-the-art VPR methods, including seven one-stage approaches based on global feature retrieval, similar to ours: NetVLAD [2], SFRS [27], CosPlace [16], MixVPR [18], EigenPlaces [17], SALAD [20], and FoL-global [10], as well as two two-stage methods with re-ranking: TransVPR [9] and SelaVPR [1]. Notably, MixVPR, SALAD, and FoL-global are trained on the same dataset, GSV-Cities. In contrast, CosPlace and EigenPlaces utilize purpose-built large-scale datasets (SF-XL). We follow SelaVPR++ [28] using multiple VPR datasets (including Pitts30k-train, MSLS-train, SF-XL, and GSV-Cities) for training. Furthermore, recent methods such as SelaVPR, SALAD, and FoL-global employ the foundation model DINOv2 as the backbone (SelaVPR and FoL-global using DINOv2-large, and the others using DINOv2-base) achieving state-of-the-art performance on the VPR task. Following this trend, we adopt DINOv2-large as our backbone. As shown in Table 1, our 3GeM Pooling achieves the best

performance across nearly all evaluated datasets.

Methods based on the DINOv2 backbone—including SelaVPR, SALAD, FoL-global, and our proposed 3GeM Pooling—consistently demonstrate strong performance, outperforming other approaches across datasets with diverse and challenging conditions. Each of these methods fine-tunes DINOv2 in distinct ways, highlighting that the powerful feature representation capability of DINOv2, when combined with appropriate fine-tuning strategies, is sufficient to address most challenges in VPR. Notably, our method adopts a more compact feature representation than SALAD, and FoL-global, yet achieves superior results on all three benchmark datasets. Specifically, it attains 83.7% R@1 on the MSLS-challenge dataset, significantly surpassing other global-retrieval-based and two-stage methods. This demonstrates the strong robustness of our method in coping with diverse visual condition variations, including low image quality, significant scene depth, and long-term scene changes. Importantly, our approach achieves the best performance across all three datasets while using the lowest global descriptor dimensionality—only 1024 dimensions—considerably fewer than those employed by other methods, which often use feature dimensions several times larger.

Fig. 3 presents qualitative results showing the superiority of 3GeM in challenging scenarios. The selected examples involve various difficulties, including severe condition variations, extreme viewpoint changes, and occlusions caused by dynamic objects. While the two competing methods return visually similar but incorrect images indicating a vulnerability to perceptual aliasing our 3GeM consistently retrieves the correct reference images, demonstrating strong robustness against these challenges. In particular, our method is highly effective in handling extreme viewpoint variations.

4.4. Ablation Study

Effect of 3GeM Pooling layer. We first evaluate the effectiveness of 3GeM Pooling layer by comparing it to the standard GeM Pooling. To be fair, we use a fully connected layer after both pooling layers to unify the dimensions of the global descriptors. The backbone of both pooling layers uses DINOv2-base [19]. We train on two training sets (Pitts30k and GSV-Cities), and test on two test sets (Pitts30k-test and MSLS-val). The results are shown in Table 2. Our experiments demonstrate that 3GeM Pooling consistently outperforms GeM Pooling across most settings. In particular, on the Pitts30k-test set, 3GeM achieves an improvement of up to 6.0% in R@1 compared to GeM. Notably, the performance gain is more pronounced when training data is limited, highlighting the effectiveness

Table 1. Comparison to state-of-the-art methods on four VPR benchmarks. Best in **bold**, second underlined. Two-stage method dims omitted.

Method	Backbone	Dim.	Pitts30k			MSLS-val			MSLS-challenge			Nordland		
			R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
NetVLAD [2]	VGG16	32768	81.9	91.2	93.7	53.1	66.5	71.1	35.1	47.4	51.7	6.4	10.1	12.5
SFRS [27]	VGG16	4096	89.4	94.7	95.9	69.2	80.3	83.1	41.6	52.0	56.3	16.1	23.9	28.4
TransVPR [9]	/	/	89.0	94.9	96.2	86.8	91.2	92.4	63.9	74.0	77.5	63.5	68.5	70.2
CosPlace [16]	VGG16	512	88.4	94.5	95.7	82.8	89.7	92.0	61.4	72.0	76.6	58.5	73.7	79.4
MixVPR [18]	ResNet50	4096	91.5	95.5	96.3	88.0	92.7	94.6	64.0	75.9	80.6	76.2	86.9	90.3
EigenPlaces [17]	ResNet50	2048	92.5	96.8	97.6	89.1	93.8	95.0	67.4	77.1	81.7	71.2	83.8	88.1
SelaVPR [1]	DINOv2-large	/	92.8	96.8	97.7	90.8	96.4	97.2	73.5	87.5	90.6	87.3	93.8	95.6
SALAD [20]	DINOv2-base	8448	92.5	96.4	97.5	92.2	96.4	97.0	75.0	88.8	91.3	<u>89.7</u>	95.5	97.0
FoL-global [10]	DINOv2-large	8848	93.9	97.2	98.1	<u>93.1</u>	<u>96.9</u>	<u>97.4</u>	<u>78.7</u>	<u>90.8</u>	<u>93.0</u>	87.8	96.4	97.7
3GeM (ours)	DINOv2-large	1024	<u>93.5</u>	<u>96.9</u>	<u>97.9</u>	93.9	97.3	98.0	83.7	92.8	94.6	94.1	98.1	98.9

**Fig. 3.** Qualitative results on four challenging cases (view-point/condition changes, dynamic objects, etc.). Our 3GeM retrieves the correct references, while others fail.**Table 2.** Ablation Study of 3GeM and SGCE Modules (In the method column, (P) denotes training on Pitts30k, and (G) denotes training on GSV-Cities). Backbone: DINOv2-base.

Method	Pitts30k			MSLS		
	R@1	R@5	R@10	R@1	R@5	R@10
GeM+fc (P)	78.5	89.8	92.7	53.8	70.0	<u>78.9</u>
3GeM+fc (P)	<u>84.5</u>	<u>92.2</u>	<u>94.5</u>	<u>61.6</u>	<u>73.1</u>	77.8
3GeM+SGCE (P)	84.6	92.8	95.0	69.2	82.0	84.7
GeM+fc (G)	90.2	95.3	96.6	86.4	94.1	95.4
3GeM+fc (G)	<u>90.6</u>	<u>95.5</u>	<u>96.7</u>	<u>88.8</u>	<u>94.5</u>	<u>95.5</u>
3GeM+SGCE (G)	92.1	96.3	97.5	91.5	95.9	96.4

of 3GeM in low-data regimes. It is also worth noting the differences in data characteristics: Pitts30k primarily consists of urban street view images, whereas MSLS includes a broader variety of scenes, including urban, rural, and natural environments. Thus, training on Pitts30k and testing on MSLS offers a good indication of a model's generalization ability. With Pitts30k training, 3GeM boosts MSLS-val R@1 from 53.8% (GeM) to 61.6%, indicating improved generalization.

Effect of the SGCE module. In this section, we compare the performance of the 3GeM method followed by either a fully connected (FC) layer or our proposed SGCE module. The models are trained on two datasets, Pitts30k and GSV-Cities, and evaluated on the Pitts30k and MSLS test sets. Results show that incorporating the SGCE module consistently improves performance across both

Table 3. Comparison of 768-Dimensional Global Descriptors Using DINOv2-base Trained on GSV-Cities.

Method	Pitts30k			MSLS		
	R@1	R@5	R@10	R@1	R@5	R@10
GeM Pooling	89.5	95.0	96.3	85.4	93.0	94.3
class token	91.4	96.2	97.4	88.4	95.1	96.4
1-Cluster VLAD [8]	91.6	96.2	97.4	90.4	95.3	96.4
3GeM+SGCE(ours)	92.1	96.3	97.5	91.5	95.9	96.4

datasets, as shown in Table 2. Notably, when trained on Pitts30k and tested on MSLS, the SGCE-enhanced model achieves a significant improvement of +7.6% in R@1 and +8.9% in R@5. These results demonstrate that the proposed SGCE module, which integrates spatial features into channel-wise representations, enhances the model's discriminative power, robustness, and generalization ability. We found that performance is stable across different choices of γ , with the best results around 0.55.

Comparison with other ultra-low-dimensional methods. We further compare our 3GeM Pooling method with other ultra-low-dimensional global descriptor approaches, including GeM Pooling, 1-cluster VLAD, and the class token representation. We conduct a comparison of low-dimensional global descriptors that match the dimensionality of the local descriptors. Specifically, each method outputs a 768-dimensional global descriptor based on the DINOv2-base backbone. All models are trained on the GSV-Cities dataset. The results, presented in Table 3, demonstrate that 3GeM consistently outperforms all three methods across both test sets: MSLS and Pitts30k. Some of the comparative results are taken from [8]. These findings indicate that 3GeM achieves state-of-the-art performance among ultra-low-dimensional global descriptor methods. The strong performance of 3GeM in ultra-low-dimensional settings makes it especially suitable for deployment in VPR tasks on resource-constrained edge devices.

5. CONCLUSIONS

In this paper, we proposed 3GeM Pooling, a compact yet powerful global descriptor for VPR. By extending GeM pooling to spatial and channel dimensions, 3GeM captures richer directional cues while retaining the simplicity of the original GeM. Together with the SGCE module, which refines channel features via spatial attention, our method delivers state-of-the-art performance across diverse benchmarks with only 1024 dimensions—significantly smaller than most existing descriptors. Beyond accuracy, 3GeM demonstrates strong generalization across datasets and resilience in low-data regimes, highlighting its robustness and practical utility.

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